

SYSTEM FORENSIC SCANNER (SFS)

IMPLEMENTATION DESIGN DOCUMENT

Version: 1.0

Status: Ready For Development

PART 1 - DATABASE SCHEMA DETAIL

Database:

PostgreSQL 17+

Schema:

```
CREATE SCHEMA sfs;
```

TABLE USERS

```
CREATE TABLE sfs.users (  
    id UUID PRIMARY KEY,  
    username VARCHAR(100) UNIQUE NOT NULL,  
    password_hash TEXT NOT NULL,  
    created_at TIMESTAMP NOT NULL,  
    updated_at TIMESTAMP NOT NULL  
);
```

Indexes:

```
CREATE INDEX idx_users_username  
ON sfs.users(username);
```

TABLE PROJECTS

```
CREATE TABLE sfs.projects (  
    id UUID PRIMARY KEY,
```

```
name VARCHAR(255) NOT NULL,  
  
target_type VARCHAR(100) NOT NULL,  
  
target_path TEXT NOT NULL,  
  
baseline_path TEXT NOT NULL,  
  
description TEXT,  
  
created_at TIMESTAMP NOT NULL,  
  
updated_at TIMESTAMP NOT NULL  
);
```

TABLE SCANS

```
CREATE TABLE sfs.scans (  
  id UUID PRIMARY KEY,  
  
  project_id UUID NOT NULL,  
  
  status VARCHAR(50) NOT NULL,  
  
  score INTEGER,  
  
  verdict VARCHAR(50),  
  
  started_at TIMESTAMP,  
  
  finished_at TIMESTAMP,  
  
  created_at TIMESTAMP NOT NULL  
);
```

Foreign Key:

```
ALTER TABLE sfs.scans  
ADD CONSTRAINT fk_scans_project  
FOREIGN KEY(project_id)  
REFERENCES sfs.projects(id);
```

TABLE FINDINGS

```
CREATE TABLE sfs.findings (  
  id UUID PRIMARY KEY,  
  
  scan_id UUID NOT NULL,  
  
  module VARCHAR(100),  
  
  severity VARCHAR(50),  
  
  title TEXT,  
  
  description TEXT,  
  
  evidence JSONB,  
  
  created_at TIMESTAMP NOT NULL  
);
```

Indexes:

```
CREATE INDEX idx_findings_scan  
ON sfs.findings(scan_id);  
  
CREATE INDEX idx_findings_severity  
ON sfs.findings(severity);
```

TABLE EVIDENCE

```
CREATE TABLE sfs.evidence (  
  id UUID PRIMARY KEY,  
  
  scan_id UUID NOT NULL,  
  
  category VARCHAR(100),  
  
  path TEXT,  
  
  metadata JSONB,  
  
  created_at TIMESTAMP  
);
```

Categories:

```
hash
diff
yara
ioc
katana
nuclei
httpx
database
timeline
```

TABLE TIMELINE

```
CREATE TABLE sfs.timeline (
  id UUID PRIMARY KEY,

  scan_id UUID NOT NULL,

  source VARCHAR(100),

  event_time TIMESTAMP,

  title TEXT,

  details JSONB
);
```

TABLE REPORTS

```
CREATE TABLE sfs.reports (
  id UUID PRIMARY KEY,

  scan_id UUID NOT NULL,

  json_path TEXT,

  html_path TEXT,

  pdf_path TEXT,

  generated_at TIMESTAMP
);
```

TABLE BASELINES

```
CREATE TABLE sfs.baselines (  
  id UUID PRIMARY KEY,  
  
  project_id UUID NOT NULL,  
  
  name VARCHAR(255),  
  
  source_path TEXT,  
  
  version VARCHAR(100),  
  
  created_at TIMESTAMP  
);
```

TABLE FILE_HASHES

```
CREATE TABLE sfs.file_hashes (  
  id UUID PRIMARY KEY,  
  
  scan_id UUID NOT NULL,  
  
  file_path TEXT NOT NULL,  
  
  sha256_hash TEXT NOT NULL,  
  
  file_size BIGINT,  
  
  modified_time TIMESTAMP  
);
```

PART 2 - OPENAPI SPECIFICATION

Base URL

/api/v1

Content-Type

application/json

Authentication

Bearer Token

AUTH

Login

POST

`/api/v1/auth/login`

Request

```
{
  "username": "admin",
  "password": "password"
}
```

Response

```
{
  "token": "jwt-token"
}
```

PROJECTS

Create Project

POST

`/api/v1/projects`

Request

```
{
  "name": "OJS Production",
  "targetType": "ojs",
  "targetPath": "D:/ojs",
}
```

```
{
  "baselinePath": "D:/ojs-clean"
}
```

List Projects

GET

```
/api/v1/projects
```

Get Project

GET

```
/api/v1/projects/{id}
```

Delete Project

DELETE

```
/api/v1/projects/{id}
```

SCANS

Start Scan

POST

```
/api/v1/scans/start
```

Request

```
{
  "projectId": "uuid"
}
```

Response

```
{
  "scanId": "uuid",
  "status": "running"
}
```

Scan Status

GET

```
/api/v1/scans/{id}
```

Scan Findings

GET

```
/api/v1/scans/{id}/findings
```

Scan Timeline

GET

```
/api/v1/scans/{id}/timeline
```

Scan Report

GET

```
/api/v1/scans/{id}/report
```

BASELINES

Upload Baseline

POST

`/api/v1/baselines`

List Baselines

GET

`/api/v1/baselines`

REPORTS

Download JSON

GET

`/api/v1/reports/{id}/json`

Download HTML

GET

`/api/v1/reports/{id}/html`

Download PDF

GET

`/api/v1/reports/{id}/pdf`

PART 3 - SPRINT BREAKDOWN

Sprint 1

Duration

2 minggu

Goal

Project Foundation

Tasks

- Setup Go Workspace
- Setup PostgreSQL
- Setup Gin
- Setup GORM
- Authentication
- Project CRUD
- Baseline CRUD

Deliverables

Login
Dashboard
Project Management

Sprint 2

Goal

Core File Forensics

Tasks

- Hash Engine
- Baseline Compare
- Diff Engine
- File Audit Engine

Deliverables

Added File Detection
Modified File Detection
Deleted File Detection

Sprint 3

Goal

Malware Detection

Tasks

- YARA Integration
- IOC Engine
- Signature Loader
- Local Cache

Deliverables

Malware Detection
Webshell Detection
IOC Detection

Sprint 4

Goal

Database Forensics

Tasks

- SQL Parser
- SQL.GZ Parser
- MariaDB Support

Detection

- New Users
- New Admins
- Suspicious Roles

Sprint 5

Goal

Attack Surface Discovery

Tasks

- Katana Integration
- Httpx Integration

Output

- Endpoints
 - Parameters
 - Upload Pages
 - Admin Pages
-

Sprint 6

Goal

Vulnerability Assessment

Tasks

- Nuclei Integration
- Custom Templates
- OJS Templates
- Wordpress Templates

Deliverables

CVE Detection
Misconfiguration Detection

Sprint 7

Goal

Timeline Reconstruction

Tasks

- Apache Parser
- Nginx Parser
- Application Log Parser
- Timeline Builder

Deliverables

Incident Timeline

Sprint 8

Goal

Reporting

Tasks

- JSON Report
- HTML Report
- PDF Report
- Risk Engine

Deliverables

Executive Report
Technical Report

Sprint 9

Goal

Production Hardening

Tasks

- Benchmark
- Memory Optimization
- Parallel Workers
- Security Review
- Testing

Target

100.000 files
< 15 menit

DEVELOPMENT PRIORITY

Priority 1

Hash
Baseline
Diff

Priority 2

YARA
IOC

Priority 3

Database Forensics

Priority 4

Katana
Httpx
Nuclei

Priority 5

Timeline
Reports

RECOMMENDED MVP RELEASE

Version:

v0.1.0

Modules Included:

✓ Baseline Engine

✓ Hash Engine

✓ Diff Engine

✓ YARA Engine

✓ IOC Engine

✓ PostgreSQL

✓ JSON Report

Karena kombinasi modul tersebut sudah mampu menemukan sebagian besar webshell, backdoor, file modifikasi, dan persistence mechanism pada OJS, WordPress, Laravel, maupun aplikasi PHP generik sebelum Katana dan Nuclei ditambahkan pada versi berikutnya.